

Electricity Demand Forecasting in Brazil using Multi-Layer Perceptron Neural Networks

Optimizing Grid Load Management for Sustainable Energy
Distribution in the North Region

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PROJECT OVERVIEW

RESUME

Implementation of an end-to-end machine learning pipeline designed to predict hourly electricity consumption in northern Brazil.

The MLPRegressor algorithm (multilayer perceptron) was selected due to its excellent ability to map highly complex and nonlinear temporal relationships without the heavy computational load of traditional recurrent models.

BACKGROUND

Brazil's Northern electrical subsystem is uniquely isolated, yet critical due to massive hydro potentials. Grid operators (ONS) face volatile load spikes. Accurately predicting sub-daily load behavior is essential to resolve grid vulnerabilities, prevent localized brownouts, and optimize generation schedules.

IMPORTANCE

- Grid Resilience: Proactive capacity management blocks cascading transformer overloads.
- Spinning Reserves: Optimizes startup scheduling for thermal backup turbines.
- Decarbonization: Reduces carbon-heavy peak-reserve activation.

OBJECTIVES

- Ingest, clean, and chronologically sort 46,320 records of ONS historical hourly data.
- Scale metrics dynamically to minimize training gradient explosion.
- Forecast next-hour demand cycles, limiting error standard deviation under 2%.

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METHODOLOGY

DATA INGESTION & SCALE PREP

- ONS Historical Feed: Ingested 46,320 records from the authentic SamuelM0422/Hourly-Electricity-Demand-Brazil-Dataset on Hugging Face.
- NORTE Isolation: Extracted load targets exclusively using the `val_cargaenergiahomwmed` parameter, discarding unrelated regional grids.
- Min-Max Scaling: Scaled values to the [0, 1] interval to boost SGD convergence rates and prevent saturation of the activation nodes.
- Min-Max Normalization:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

LAG ENGINEERING & DATA SPLITS

- Time Series Lag Windowing: To capture sequential temporal dependencies without complex recurrent neural layer overhead:
 - 24-Hour Sliding Horizon: Vector utilizes inputs from T_{-24} to T_{-1} sequentially to resolve the targeted target value at T_0 .
 - Feature Dimensions: Generated 46,296 vectors, preserving historical alignment.
- Chronological Data Splits: To prevent data leakage during optimization, we implemented strict sequential timeline slicing:
 - Training Set (80%): Covers 37,036 samples to lock-in multi-season trends.
 - Testing Set (20%): Covers 9,260 validation samples reserved exclusively for out-of-sample prediction metrics.

DEEP MLPREGRESSOR ARCHITECTURE

- Hidden Layer Topology: Two dense feedforward hidden layers with (100, 50) nodes respectively, yielding robust representational capacity.
- Activation Function: Rectified Linear Unit (ReLU) applied at hidden nodes to eliminate gradient saturation during backpropagation.
- Solver Optimization: Trained via Adam (Stochastic Gradient-Based Optimizer) for adaptive step-size mapping.
- Convergence Rules: Iterations bounded to a 500 max epoch threshold with early stopping enabled.

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RESULTS AND ANALYSIS

MODEL DIAGNOSTIC PERFORMANCE

1.72%

Normalized Error STD. DEV

182.13

RMSE (MWmed)

136.24

Standard Deviation of Error

Root Mean Squared Error

Root Mean Squared Error (RMSE):

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Interpretation for Decision-Making and Grid Operations

- 98.28% Base Accuracy: The standardized error of 0.0172 indicates that the model’s discrepancies account for just 1.72% of the region’s average consumption, a benchmark of excellence in the energy industry (where the target is typically < 5%).
- Excellent Peak Tracking (RMSE: 182.13): Indicates the average hourly margin of error. The model closely tracks sudden spikes in demand during peak hours without lagging.
- Predictable Errors with No Surprises (Std Dev Error: 136.24): As this is a low and controlled deviation, it confirms that the model’s errors are stable and uniform. There are no drastic variations or random catastrophic errors, allowing for the establishment of highly precise energy reserve margins.

Projected 48-Hour Load Schedule

Interval Key	Day 1 (2025-04-14) predicted Load	Day 2 (2025-04-15) predicted Load	Subsystem Trend Outlook
00:00 (Midnight Peak)	8,240.47 kWh	7,336.23 kWh	Fast drop-off past peak interval
06:00 (Morning Valley)	7,098.96 kWh	6,790.79 kWh	Baseline load stabilization
12:00 (Midday Base)	6,784.79 kWh	6,175.56 kWh	Gradual industrial energy cool down
23:00 (Night Valley)	7,228.98 kWh	6,532.20 kWh	Climbing towards next peak phase

DIAGNOSTICS HIGHLIGHT

Although the forecast is generated recursively (the model uses its own past predictions to build the windows for the coming hours), the curve remains balanced. This demonstrates that the pipeline preserves the logic of daily consumption cycles without becoming distorted or diverging as time passes.

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CONCLUSIONS

Technical Validity

The dual-layer MLP network successfully learned underlying circadian rhythms without training-stage overfitting, demonstrating predictive reliability across cyclic variations.

Grid Management

Equipped with a recursive 48-hour load pipeline, dispatch operators can dynamically scale reserve turbines and limit transmission congestion bottlenecks.

Scalability

Future pipelines can easily integrate meteorological data (temperature/humidity) to account for seasonal heatwave consumption surges.

Error Stability and Risk Mitigation

The tightly bounded error distribution ($\text{Error Std. Dev: } 136.24 \text{ MWmed}$) confirms that prediction deviations are highly centralized and uniform. By eliminating systemic drift and extreme outliers, the model effectively mitigates the risk of sudden, unpredicted load deficits, ensuring a dependable safety margin for grid operations.

Data Pipeline Optimization and Efficiency

By substituting complex recurrent architectures with a streamlined 24-hour lag windowing strategy, the project achieves an optimal balance between computational efficiency and accuracy. This architecture delivers high-precision, sub-daily forecasting while maintaining low training overhead, making it highly suitable for near-real-time production deployment.

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Variable / Feature	Data Type	Role in the Model	Technical Description and Impact
`id_subsistema` / `nom_subsistema`	Text (Categorical)	intake filter	Identifiers of the Brazilian network regions. These were used only to segment the data for the «NORTE ("N")» region and were discarded for training purposes.
`din_instante`	Date / Time	Time Axis	Original time stamp by the hour. A strict chronological order is required to ensure the division and integrity of the time series.
`val_cargaenergiahomwmed`	Floating (Numeric)	Original Goal	Actual load recorded in megawatts (average). This is the target variable to be modeled and projected.
`demanda_norm`	Floating (0 a 1)	Scaled Goal	Result of applying a linear transformation of minimum and maximum values to energy consumption to stabilize the neurons.
`Demanda_t-1` to `Demanda_t-24`	Floating (0 a 1)	Predictive Variables	The 24 lag variables created dynamically. They represent the lookback window used by the network to predict the next time step.